

SQL-Practice

For freesql.com

Your available tables are:

- `HR.EMPLOYEES`
- `HR.DEPARTMENTS`
- `HR.JOBS`
- `HR.JOB_HISTORY`
- `HR.LOCATIONS`
- `HR.COUNTRIES`
- `HR.REGIONS`

So now I'll give you lab-style questions based exactly on these Oracle HR tables, with the same level as your college practicals.

Practice Set — Oracle HR Schema

Basic SELECT Queries

1. Display all employees working in department 60.
 2. Show first name, last name, and salary of employees earning more than 8000.
 3. Display employee id, first name, and manager id.
 4. Show employees whose job id is `IT_PROG`.
 5. Display employees whose salary is between 5000 and 10000.
 6. Show employees whose first name starts with `S`.
 7. Display employees whose last name contains the letter `a`.
 8. Show employee names and salaries in descending order of salary.
 9. Display employee name and hire date sorted by latest hire date.
 10. Show all distinct job ids from employees table.
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String & Pattern Queries

1. Display employees whose first name has exactly 5 characters.
2. Show employees whose email ends with `N`.
3. Display employee first name and annual salary (`salary * 12`).
4. Show employee names in uppercase.

5. Display first 3 characters of employee first names.
 6. Concatenate first name and last name as full name.
 7. Find employees whose phone number contains 124.
 8. Display length of employee first names.
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Aggregate Functions

1. Find total salary department-wise.
 2. Display average salary for each job id.
 3. Find maximum and minimum salary in each department.
 4. Count employees department-wise.
 5. Show departments having more than 5 employees.
 6. Display job ids whose average salary is greater than 10000.
 7. Find highest salary in each job role.
 8. Display total salary expenditure for each department.
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Subqueries

1. Find employees earning more than average salary.
 2. Display employees working in the same department as employee 'Lex'.
 3. Find employees getting the maximum salary.
 4. Show employees who are not managers.
 5. Find employees earning second highest salary.
 6. Display departments having no employees.
 7. Find employees whose salary is greater than all employees in department 50.
 8. Show employees hired after 'Neena'.
 9. Display employees working in department with highest average salary.
 10. Find employees earning less than department average salary.
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Join Queries

1. Display employee names along with department names.
2. Show employee name and manager name using self join.
3. Display department name and city.
4. Show employees working in departments located in 'Seattle'.
5. Display country name with region name.
6. Show employees and their job titles.

7. Display employees with department name and location city.
 8. Show all departments even if no employees exist.
 9. Display employees who do not belong to any department.
 10. Show job history along with employee names.
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Group By + Having

1. Count employees job-wise.
 2. Display departments where total salary exceeds 50000.
 3. Show jobs having average salary greater than 9000.
 4. Find departments with more than 3 employees.
 5. Display highest salary department-wise.
 6. Show average salary location-wise.
 7. Count employees country-wise.
 8. Find regions having more than 2 countries.
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DDL & Constraints

1. Create a table named `student`.
 2. Add `NOT NULL` constraint on student name.
 3. Add `UNIQUE` constraint on email column.
 4. Add `CHECK` constraint so marks are between 0 and 100.
 5. Add a column `phone_number`.
 6. Modify size of a varchar column.
 7. Rename a column.
 8. Drop a constraint.
 9. Drop the table.
 10. Create a table with primary key and foreign key.
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PL/SQL Practice

1. Print numbers from 1 to 10 using loop.
2. Find factorial of a number.
3. Check whether a number is even or odd.
4. Increase employee salary by 10% if salary < 7000.
5. Accept employee id and display salary.
6. Count total employees in department 50.

7. Display maximum salary from employees table.
 8. Print multiplication table of 7.
 9. Reverse a number.
 10. Check palindrome number.
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Cursor Practice

1. Create cursor to display all employee names.
 2. Use implicit cursor to update salary.
 3. Display employees department-wise using cursor.
 4. Print all employees having salary greater than 10000 using cursor.
 5. Count employees using explicit cursor.
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Trigger Practice

1. Create audit table for employee updates.
 2. Create trigger to store deleted employee records.
 3. Prevent insertion of negative salary using trigger.
 4. Automatically update modified date using trigger.
 5. Create trigger before delete on departments table.
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Advanced Practice

1. Find top 3 highest paid employees.
2. Rank employees based on salary.
3. Display running total of salaries.
4. Find duplicate salaries.
5. Show employees hired in current year.
6. Display department with highest total salary.
7. Find employees whose salary is above department average.
8. Display employees working in same city as their manager.
9. Find employees with no job history.
10. Show latest hired employee department-wise.